

DRAFT - INSERT EXACTLY ONE SUCCESSFULLY GRADED PHASE-II SUBMISSION ID BEFORE PRIZE FILING

Moment-Tangent Control for Sparse Randomized-QMC Mean Estimation of Deep ReLU Networks

Frozen Phase-II candidate and Algorithmic Contribution Prize report draft
Team: gmmrk | Frozen: 2026-08-03 | Evaluator: WHestBench 0.14.0 / FlopScope 0.10.0

Primary development result	Frozen lockbox result	Artifact result
17.1599% nested OOF raw-MSE reduction	9.91146e-7 raw MSE on untouched n=200	10/10 cold starts; 0 isolated failures

Executive conclusion

The promoted estimator is a rules-compliant, fully offline randomized-QMC algorithm for the width-256, depth-32 bias-free ReLU networks used by WHestBench Phase II. Its material innovation is a fixed moment-tangent control: first-layer empirical mean and variance residuals are propagated through a deterministic mean-field response map and subtracted from the final randomized-QMC estimate with development-frozen coefficient $\lambda = 0.9807112198896164$.

The control reduced nested whole-network out-of-fold raw MSE by 17.1599% on 800 public development networks. The effect repeated across three independent setup seeds and improved the official 20-network adjusted score by 19.8094% with zero failures. After source and archive freeze, one untouched 200-network lockbox produced 9.91146e-7 raw MSE and passed every predeclared mean, tail, and finiteness gate. No lockbox-driven retuning followed.

This report does not claim a guaranteed prize or private-rerun score. The complete estimator is deliberately described as biased because its pilot-rescued sparsification can omit active upstream paths. The moment-tangent control is centered under an ideal continuous-Gaussian law, with finite-grid and floating-point qualifications.

1. Challenge objective and accounting

For M networks, WHestBench ranks the mean final-layer error after a compute multiplier. With budget B = 272e9 FLOPs and effective compute C_m including charged residual wall time:

score = (1/M) sum_m MSE_m * max(0.1, C_m / B)

All MLP-dependent numerical work in the submitted source uses FlopScope arrays or flops.stats. The estimator does not call an API, network service, subprocess, raw NumPy prediction path, grader state, or an accounting bypass. Setup reads one packaged asset from SetupContext.submission_dir and randomizes it only with the official setup seed.

2. Frozen estimator

14,000-point Sobol prefix	digital shift + Box-Muller	antipodes + radial weights	cold-screen + pilot rescue	moment tangent lambda=0.980711
---------------------------	----------------------------	----------------------------	----------------------------	--------------------------------

Randomized low-discrepancy paths

The packaged asset contains 32,768 points in 256 dimensions from a fixed LMS+shift-scrambled Sobol construction. Prediction uses its first 14,000 rows. Setup applies an independent uint32 XOR shift from the official seed, maps midpoints to approximate standard normals with Box-Muller, and propagates each path with its antipode. The final submission therefore evaluates 28,000 paths before pruning.

Exact spectral radial control

A bias-free ReLU network is positively one-homogeneous: $f(r\ u) = r\ f(u)$. For $X = R\ U$ with R chi-distributed in dimension 256, the following final-layer weight preserves the ideal Gaussian target while changing variance:

$$w(R) = 1 - (2600/537689)(R^2 - 257) + (3/537689)(R^4 - 66563)$$

The identities $E[R(R^2-257)] = 0$ and $E[R(R^4-66563)] = 0$ make this an exact ideal-law radial control for every final neuron. A separate spherical-radial conditioning mutation was tested at full scale and rejected because its 0.1859% mean raw gain was not statistically stable.

Pilot-rescued sparsification

A diagonal-Gaussian analytic pass computes layer means, variances, standardized preactivation alpha, and firing probabilities. Neurons with alpha below -2 are provisionally cold. A 256-antipodal-pair pilot rescues any such neuron observed firing. Dense propagation continues only through the active set; analytically cold final outputs are filled by mean propagation.

This is an empirical rescue heuristic, not a proof of zero contribution. A neuron can evade the pilot while carrying an omitted upstream path. That bias is retained explicitly in every claim and artifact.

3. Moment-tangent control

For first-layer neuron j with weight norm σ_j and analytic mean $\mu_j = \sigma_j / \sqrt{2\ \pi}$, antithetic paths provide empirical first and second moments. Define centered residuals:

$$\begin{aligned} r_{\mu} &= \text{mean}(h_1) - \mu \\ r_v &= [\text{mean}(h_1^2) - \sigma^2/2] - 2\ \mu\ r_{\mu} \end{aligned}$$

The ideal Gaussian expectations of these residuals are zero. A deterministic response approximation propagates them through each later layer. For preactivation perturbations δ_m and δ_v at standardized mean alpha with analytic ReLU mean μ_{plus} :

$$\begin{aligned} \delta_{\mu\text{plus}} &= \Phi(\alpha)\ \delta_m + \phi(\alpha)\ \delta_v / (2\ \sigma) \\ \delta_{v\text{plus}} &= 2\ \mu_{\text{plus}}\ \delta_m + \Phi(\alpha)\ \delta_v - 2\ \mu_{\text{plus}}\ \delta_{\mu\text{plus}} \end{aligned}$$

The final correction is $Q_{\text{corrected}} = Q_{\text{base}} - \lambda * \delta_{\mu\text{plus_final}}$. Because the propagation map and coefficient depend only on network weights and frozen development constants, response-approximation error changes variance under the ideal law but does not create ideal-law mean shift. The fitted coefficient is close to the mechanistic value one and remained stable across whole-network folds.

Qualification: randomized uint32 midpoint Box-Muller is a finite floating-point law, not literal continuous Gaussian measure. The complete pilot-pruned estimator is biased independently of this microscopic control-centering issue.

4. Experimental firewall

Stage	Networks	Permitted use	Status
Legacy mini	100	Early premise work only; known reused data	Not a holdout
Development	0..799	Nested CV, coefficients, ablations, seed tests	Used for generation
Public lockbox	800..999	One frozen estimator, once, no fitting	Opened after hash freeze
Private Phase II	Fresh hidden	Organizer rerun of one designated submission	Remaining decisive test

The 1,000-network public dataset was partitioned and timestamp-frozen before any full-dataset score. Analysis sliced indices 0..799 before residual or score formation. Candidate source, constants, official tarball, and SHA-256 hashes were frozen before the one-shot lockbox capture. Lockbox acceptance limits were also written before access. No model mutation followed the result.

5. Evidence

Gate	n	Metric	Result	Uncertainty / failures
Nested whole-network OOF	800	Raw MSE	1.19984e-6 -> 9.93947e-7 (-17.1599%)	paired 95% CI [-2.6591e-7, -1.5156e-7]
Frozen seed 0	200	Raw MSE change	-14.54%	CI wholly favorable
Frozen seed 1	200	Raw MSE change	-19.44%	CI wholly favorable
Frozen seed 2	200	Raw MSE change	-16.31%	CI wholly favorable
Official subprocess dev20	20	Adjusted score	3.46033e-7 -> 2.77486e-7 (-19.8094%)	0 failures
Frozen public lockbox	200	Raw MSE	9.91146e-7	finite; q95 2.81997e-6; max 9.48400e-6

Matched factorial under WHestBench 0.14

Candidate	Raw MSE	Adjusted score	Mean effective compute	Decision
Pilot-rescued base	1.06892e-6	3.46033e-7	89.651B	Reference
Moment tangent	8.47386e-7	2.77486e-7	90.199B	Promote
Trace-free rank-4 H2	1.06502e-6	3.48607e-7	90.623B	Kill
Tangent + H2	8.50230e-7	2.81341e-7	91.492B	Kill

The tangent cell improved raw MSE by 20.7249% and adjusted score by 19.8094% for only 0.6114% more effective compute on the matched dev20 panel. H2 did not pay for its compute and interacted adversely with tangent.

6. Negative results kept in the ledger

Mutation	Classical translation	Measured falsifier	Decision
Tau folding	Base-two b-adic tent transform	+71% raw MSE on matched premise gate	Killed
Fractal prefix	Dyadic/Haar prefix detail	Only about 0.4 point exploratory gain; weaker bias proof	Killed
Rank-4 H2	Price/Hermite angular response control	+0.74% adjusted alone; +1.39% after tangent	Killed
Homeostatic mosaic	Weight-only network-dependent tangent gain	+0.403% nested OOF; CI crossed parity	Killed
Spherical-radial	Exact ideal-law radius integration	-0.186% raw but CI [-6.32e-9, 2.66e-9]	Killed
Memristive state	Hysteretic/adaptive continuation	Order dependence without an exact telescoping identity	Rejected

The retinal, biological-patterning, quantum-pigment, memristive, fractal, and physics prompts were translated into testable classical operators. Metaphor alone was never counted as mechanism. The experiment ledger retains these failures to prevent retrospective cherry-picking.

6b. The non-Gaussianity wall: certified exact Gaussian-closure measurements

The estimator's central design choice is that Gaussian structure enters only as a control subtracted from a randomized estimate (Section 3), never as the predictor itself. A companion measurement campaign (2026-08-08) quantifies why, by making the competing design - the propagated Gaussian moment closure - exact, certifying it, and scoring it as a standalone estimator.

Certified machinery

Two audited components make the closure exact rather than approximate. First, a bounded-cost bivariate provider evaluates the rectified-Gaussian pair moment $E[\text{ReLU}(X_i) \text{ReLU}(X_j)]$ and its derivatives via Owen-T/Phi2 with per-call enclosure certificates, a fixed 4,048-FLOP budget per pair, charged sign comparisons, exact strata at the rank-one and zero-variance limits, and a 12,890-case adversarial census (all contained). Second, an exact zero-order full-covariance recurrence ($\mu_0 = 0, V_0 = I; a_i = \mu W_i; C_i = W_i^T V_{i-1} W_i$; post-ReLU (μ_i, V_i) from exact Tallis truncated moments [12]) propagates the complete 256 x 256 covariance through all 32 layers, FlopScope-metered inclusively at $8.30e9$ FLOPs = 3.05% of budget. Assembly agrees with 30-digit mpmath references to worst error $2.144e-9$ and with an independent closure implementation to about $2e-16$.

Measured wall

Predictor of the depth-32 final-layer mean	Bias MSE (noise-floor subtracted)	Basis
Diagonal Gaussian closure	7.175e-4	3 He-init nets, 400k-sample MC truth
Exact full-covariance Gaussian closure	9.606e-5 (range 3.70e-5 - 1.79e-4)	same 3 nets, same MC truth
Randomized sampling lineage (reference)	about 3.09e-7 raw	public-100 official runs, matched budget

The Monte-Carlo noise floor of the measurement is $1-2e-7$ per network - 200 to 1,000 times below the measured closure bias - so the numbers above are resolved signal, not sampling noise. Reading: making the covariance exact buys a factor of about 7.5 over the diagonal closure, and the remaining factor of about 300 to the sampling family is third-and-higher-cumulant structure that no Gaussian-moment closure can represent, at any compute multiplier. Even priced at a zero-cost floor multiplier the exact closure would trail the sampling lineage by more than an order of magnitude. This measurement also corrected an internal planning estimate: an $8.76e-7$ closure-oracle ceiling carried in earlier strategy documents is refuted 46-fold by direct measurement, and every downstream projection built on it was withdrawn.

Together with Section 3, the campaign's evidence supports one quantified design principle for mean-field methods in deep-network estimation: exact Gaussian structure pays when subtracted (the moment-tangent control: -19.8094% adjusted score) and fails when predicted (closure-as-estimator: 46x outside the competitive boundary). The certified provider, recurrence, metering ledgers, and the kill-gated measurement protocol are released in the recursive-estimator-folding corpus (experiments m178_certified_phi2_owent, m179_background_archive_producer, t2_closure_score_measurement; fold-ledger records 177-180).

7. Compute, package, and legality

Gate	Result
Official validate	Pass; finite (2,4) contract output
Cold subprocess starts	10/10 pass; identical final MSE and counted FLOPs
Worst cold effective compute	0.340253 of 272B budget
Isolated extracted dev3	0 failures; maximum C/B = 0.343537
Package validation	Pass; no issues
Archive contents	estimator.py; sobol_owen_u32.npz; manifest.json
Archive size	33,332,322 bytes; below 50 MiB

One local setup timeout occurred only while two full800 research jobs saturated the workstation. After contention ended, all ten cold subprocess starts passed. Direct setup was about one second, well below the default five-second limit.

The active source contains no sys.path mutation, sibling import, raw NumPy/SciPy/Torch prediction route, network access, subprocess, ctypes, pickle, grader-state read, telemetry carrier, or hidden file dependency. The only file read is submission_dir/sobol_owen_u32.npz. SciPy appears only in the offline asset generator included with the source release.

8. Reproducibility and frozen hashes

Artifact	SHA-256
estimator.py	8428026E01224F9E48F2FD72C303100909E529370183E67F376E4623113425D8
sobol_owen_u32.npz	050339EC9966BD046B4FCF53C85240F89D2CD1F7D60C30421922203045EED0CA
official tar.gz	D2E58DF64A85121770F9D51977667882DD10CDC9496B6219F5E7D8258D8CF231
lockbox capture	F02B75D253CF865D6B5C9179F69102345974DE57F1098F3C8FF9D5E046CCD2BD

The asset generator is pinned to Python 3.12.13, NumPy 2.4.6, and SciPy 1.18.0. It regenerated the exact 33,317,968-byte NPZ and verified the frozen asset hash. The source release is MIT-licensed and includes the estimator, asset generator, asset, experiment ledger, freeze record, lockbox result, and offline Headroom patch.

9. Headroom-Recursion and AI disclosure

The gmrnk/headroom-recursion repository was adapted with an OfflineTranscriptClient. The adapter accepts strict queued JSON responses for latent, answer, and judge calls, can log prompts to JSONL, raises explicit exhaustion errors, and performs no API, SDK, login, model, or network call. Its targeted tests pass (11 tests). Generation 2 prescribed setup-seed stability, a tangent/H2 factorial, self-contained packaging, and one post-freeze lockbox opening; those gates were executed as written.

OpenAI Codex substantially assisted web and literature research, mathematics, code implementation, adversarial audits, experiment design, testing, analysis, and report writing. Multiple Codex subagents independently audited the biology translation, Hermite algebra, competition legality, and final artifact. Offline Headroom replay does not mean the overall project had no model assistance. All AI-generated claims were treated as hypotheses until checked against source, derivation, or measured evidence.

Section 6b's certified provider, recurrence, measurements, and text were produced with Anthropic Claude (Table 5) assistance under the same standard: every closure number above was generated by predeclared, kill-gated, response-free protocols with the falsifier written before the code, adversarially red-teamed by independent subagents, and checked against high-precision references before being reported.

10. Limitations and remaining private test

- The public lockbox is not the private Phase-II distribution; fresh organizer rerun remains decisive.
- Pilot-rescued sparsification is biased and can miss rare upstream paths.
- Ideal-Gaussian centering arguments are approximate under the finite uint32 and float32 implementation.
- Adjusted score depends on grader hardware through charged residual time, although measured compute has substantial headroom.
- If organizers publish a different final WhestBench or FlopScope version, all package and accounting gates must be repeated unchanged.

Final prize filing procedure: submit the frozen archive unchanged, obtain one successfully graded Phase-II submission ID, insert exactly that ID in this report, rerender and visually verify the PDF, and file it by the Algorithmic Prize deadline. Do not tune the estimator from the private or lockbox score.

References

- [1] ARC White-Box Estimation Challenge 2026. <https://www.aicrowd.com/challenges/arc-white-box-estimation-challenge-2026>
- [2] Algorithmic Contribution Prize guidelines. <https://discourse.aicrowd.com/t/algorithmic-contribution-prize-guidelines-how-arc-judges-these-prizes-discretion-technical-writeups-llm-usage/18041>
- [3] ARC townhall and private-rerun clarification. <https://discourse.aicrowd.com/t/townhall-summary-recording/18078>
- [4] WhestBench releases. <https://github.com/AIcrowd/whestbench/releases>
- [5] A. B. Owen. Scrambled net variance for integrals of smooth functions. SIAM Journal on Numerical Analysis. <https://doi.org/10.1137/S0036142994277468>
- [6] T. Goda. The b-adic symmetrization of digital nets for quasi-Monte Carlo integration. <https://arxiv.org/abs/1509.08570>
- [7] B. Poole et al. Exponential expressivity in deep neural networks through transient chaos. <https://arxiv.org/abs/1606.05340>
- [8] S. Schoenholz et al. Deep Information Propagation. <https://arxiv.org/abs/1611.01232>
- [9] R. Price. A useful theorem for nonlinear devices having Gaussian inputs. <https://doi.org/10.1109/TIT.1958.1057444>
- [10] S. Kondo and R. Asai. A reaction-diffusion wave on the skin of the marine angelfish. <https://doi.org/10.1038/376765a0>
- [11] D. B. Owen. Tables for computing bivariate normal probabilities. Annals of Mathematical Statistics 27(4), 1956. <https://doi.org/10.1214/aoms/1177728074>
- [12] G. M. Tallis. The moment generating function of the truncated multi-normal distribution. Journal of the Royal Statistical Society, Series B 23(1), 223-229, 1961. <https://doi.org/10.1111/j.2517-6161.1961.tb00408.x>

Report status: development and public-lockbox evidence complete. Prize-filing ID intentionally absent until one frozen Phase-II submission is successfully graded.